NAND TO TETRIS

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ROLL NO.: - 2501410026

Project 3

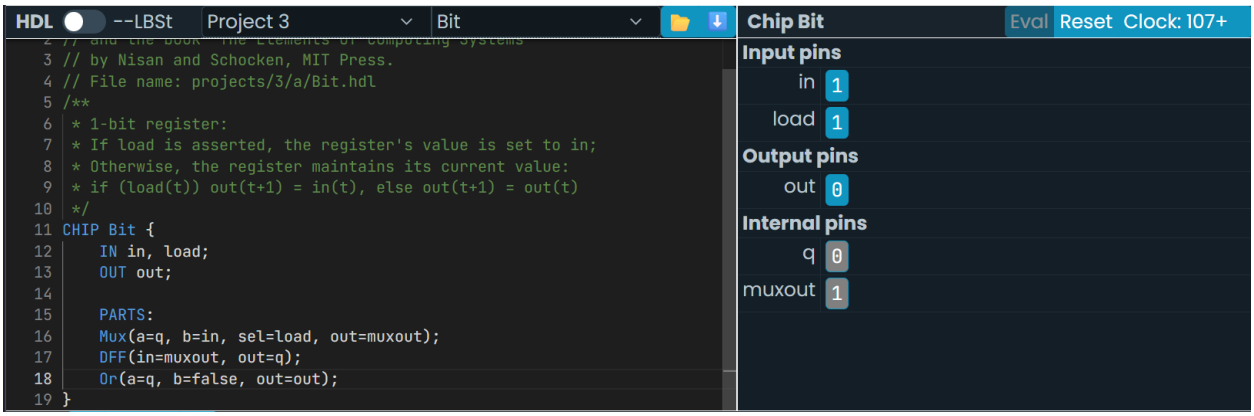
Bit

Code

```
CHIP Bit {
    IN in, load;
    OUT out;

    PARTS:
    Mux(a=q, b=in, sel=load, out=muxout);
    DFF(in=muxout, out=q);
    Or(a=q, b=false, out=out);
}
```

Screenshot



Register

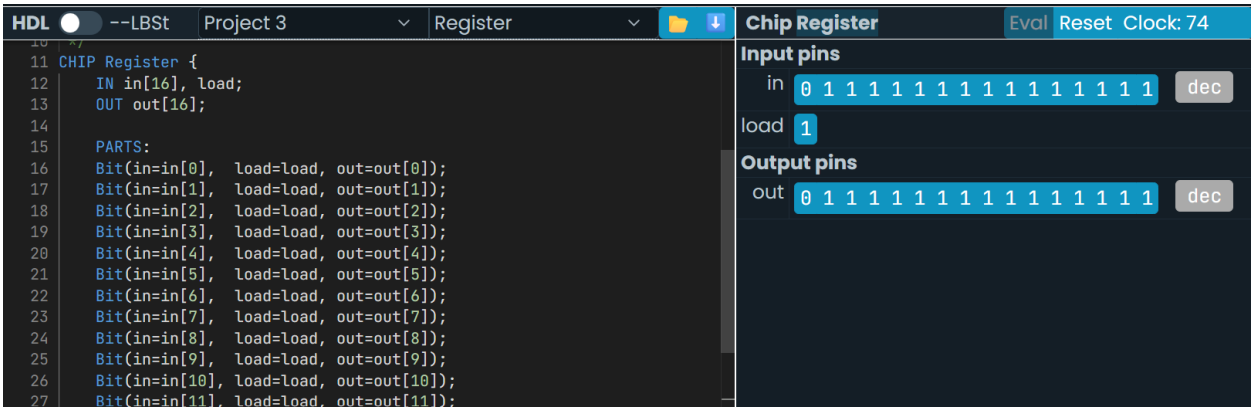
Code

```
CHIP Register {
    IN in[16], load;
    OUT out[16];

    PARTS:
    Bit(in=in[0],  load=load, out=out[0]);
    Bit(in=in[1],  load=load, out=out[1]);
    Bit(in=in[2],  load=load, out=out[2]);
    Bit(in=in[3],  load=load, out=out[3]);
    Bit(in=in[4],  load=load, out=out[4]);
    Bit(in=in[5],  load=load, out=out[5]);
    Bit(in=in[6],  load=load, out=out[6]);
    Bit(in=in[7],  load=load, out=out[7]);
    Bit(in=in[8],  load=load, out=out[8]);
    Bit(in=in[9],  load=load, out=out[9]);
```

```
    Bit(in=in[10], load=load, out=out[10]);
    Bit(in=in[11], load=load, out=out[11]);
    Bit(in=in[12], load=load, out=out[12]);
    Bit(in=in[13], load=load, out=out[13]);
    Bit(in=in[14], load=load, out=out[14]);
    Bit(in=in[15], load=load, out=out[15]);
}
```

Screenshot



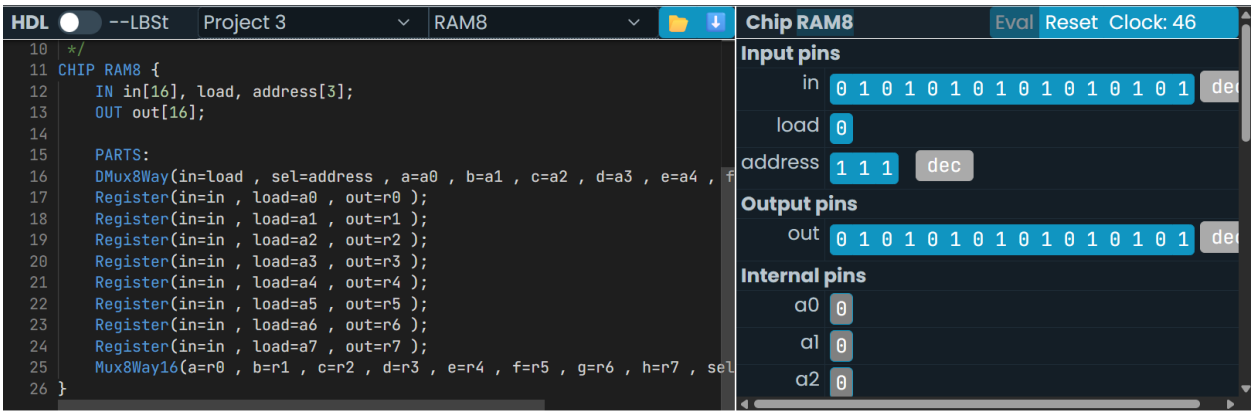
RAM8

Code

```
CHIP RAM8 {
    IN in[16], load, address[3];
    OUT out[16];

    PARTS:
    DMux8Way(in=load , sel=address , a=a0 , b=a1 , c=a2 , d=a3 , e=a4 , f=a5 , g=a6 , h=a7 );
    Register(in=in , load=a0 , out=r0 );
    Register(in=in , load=a1 , out=r1 );
    Register(in=in , load=a2 , out=r2 );
    Register(in=in , load=a3 , out=r3 );
    Register(in=in , load=a4 , out=r4 );
    Register(in=in , load=a5 , out=r5 );
    Register(in=in , load=a6 , out=r6 );
    Register(in=in , load=a7 , out=r7 );
    Mux8Way16(a=r0 , b=r1 , c=r2 , d=r3 , e=r4 , f=r5 , g=r6 , h=r7 , sel=address , out=out );
}
```

Screenshot



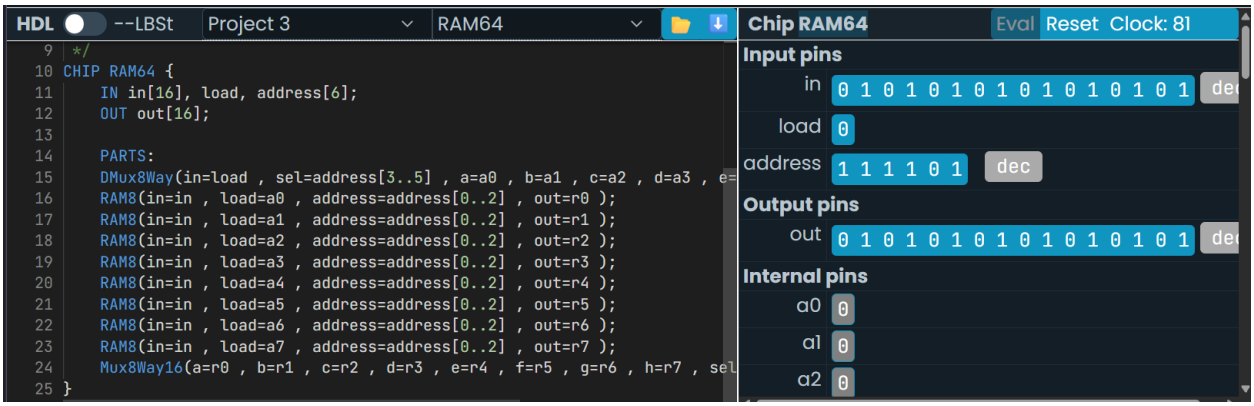
RAM64

Code

```
CHIP RAM64 {
    IN in[16], load, address[6];
    OUT out[16];

    PARTS:
    DMux8Way(in=load , sel=address[3..5] , a=a0 , b=a1 , c=a2 , d=a3 , e=a4 , f=a5 , g=a6 , h=a7 );
    RAM8(in=in , load=a0 , address=address[0..2] , out=r0 );
    RAM8(in=in , load=a1 , address=address[0..2] , out=r1 );
    RAM8(in=in , load=a2 , address=address[0..2] , out=r2 );
    RAM8(in=in , load=a3 , address=address[0..2] , out=r3 );
    RAM8(in=in , load=a4 , address=address[0..2] , out=r4 );
    RAM8(in=in , load=a5 , address=address[0..2] , out=r5 );
    RAM8(in=in , load=a6 , address=address[0..2] , out=r6 );
    RAM8(in=in , load=a7 , address=address[0..2] , out=r7 );
    Mux8Way16(a=r0 , b=r1 , c=r2 , d=r3 , e=r4 , f=r5 , g=r6 , h=r7 , sel=address[3..5] , out=out );
}
```

Screenshot



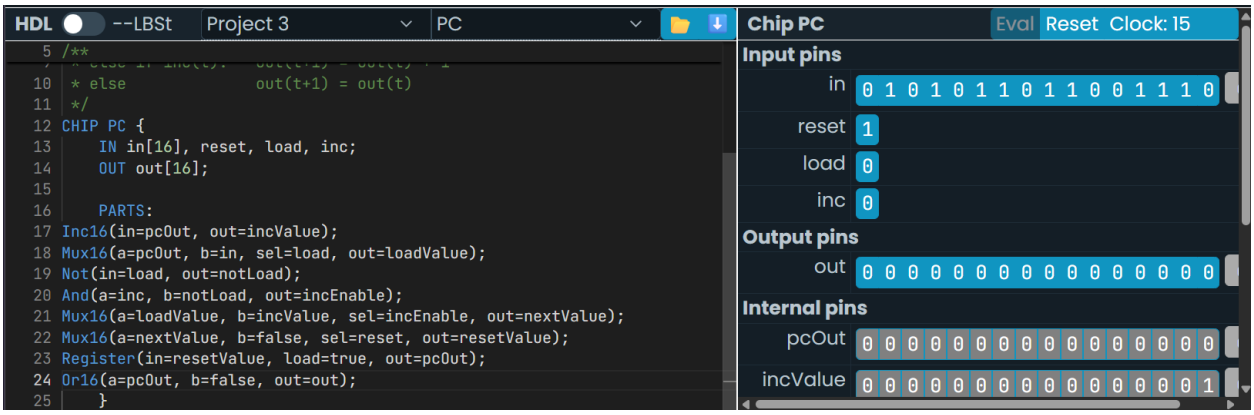
PC

Code

```
CHIP PC {
    IN in[16], reset, load, inc;
    OUT out[16];

    PARTS:
Inc16(in=pcOut, out=incValue);
Mux16(a=pcOut, b=in, sel=load, out=loadValue);
Not(in=load, out=notLoad);
And(a=inc, b=notLoad, out=incEnable);
Mux16(a=loadValue, b=incValue, sel=incEnable, out=nextValue);
Mux16(a=nextValue, b=false, sel=reset, out=resetValue);
Register(in=resetValue, load=true, out=pcOut);
Or16(a=pcOut, b=false, out=out);
    }
```

Screenshot


HDL
--LBSt
Project 3
PC
/**
* else out(t+1) = out(t)
*/
CHIP PC {
IN in[16], reset, load, inc;
OUT out[16];
PARTS:
Inc16(in=pcOut, out=incValue);
Mux16(a=pcOut, b=in, sel=load, out=loadValue);
Not(in=load, out=notLoad);
And(a=inc, b=notLoad, out=incEnable);
Mux16(a=loadValue, b=incValue, sel=incEnable, out=nextValue);
Mux16(a=nextValue, b=false, sel=reset, out=resetValue);
Register(in=resetValue, load=true, out=pcOut);
Or16(a=pcOut, b=false, out=out);
}
Chip PC
Eval
Reset
Clock: 15
Input pins
in 0 1 0 1 0 1 1 0 1 1 0 0 1 1 1 0
reset 1
load 0
inc 0
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Internal pins
pcOut 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
incValue 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1